Business Analytics - Foundations
Research & Terminology

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A. Foundational Concepts of Analytics

1. Business Analytics

The practice of using business data (the organisation's data) to analyse the performance of the business and how this can be improved as well as optimized. The data extracted through various statistical analysis, data mining, and data engineering to provide predictive insights to drive strategic and operational decision

e.g. A supermarket using sales data to increase its revenue and reduce the wasted stock.

2. Data Analytics

This is the discipline of exploring data, examining, transforming interpreting data to discover any patterns or trends and what they mean. This provides insights that support unde and decision-making.

3. Business Intelligence (BI)

Business Intelligence is a tool used to report and observe performance. it is a set of technological processes for collecting managing and analyzing the organisation's data. BI focuses o use of historical and current data to provide insights on performance of the business. It is mainly about visibility and of current and past performance

e.g. Microsoft Power BI & Tableau are popular tools ~~ut~~ used fo Business Intelligence.

4 Data - Driven Decision Making:
The process of making decisions based on data analysis (which includes collecting, cleaning, analyzing and using the data) and evidence rather than intuition, guesswork or personal experience alone.
e.g. Takealot wants to improve delivery efficiency.
- Data shows deliveries are slower in certain areas.
- Analysis reveals traffic congestion @ specific times.
- Decision - Adjust delivery routes and schedules.

5. Big Data
It is the collection and analysis of vast volumes of structured and unstructured data, generated at high speed, that require advanced technologies to store, process and extract meaningful insights. It is not just about the size of the data but rather a combination of the data volume, velocity and variety.
e.g. Netflix - generates big data from what users watch, what they skip or pause, search and preferences
(Bachman & Kosinsk, 2023)

~~6. Descriptive Analytics~~
6. Descriptive Analytics
It is the process of summarizing and interpreting historical and current data to understand past events, patterns and trends. It explains "what happened" using data. it uses data ~~a~~aggregation, collection and visualization to offer insights into business performance i.e. Tracking daily sales across stores.

7. Diagnostic Analytics
It is the process of analyzing historical datasets to discover trends, patterns, ~~and~~ relationships and the root causes. It works alongside

with descriptive analytics in answering "Why did This happen".
It focuses on understanding the causes behind past outcomes.
(Aquino.J & Jonker.A, 2023)

8. Predictive Analytics
It is using data, both historical and real-time data to answer what might happen in the future. Using historical and real-time data, forecasts of potential scenarios are formed (trend forecast) ~~(Coforma)~~ (Cote, C. 2021).
e.g. Weather forecasting predictions.

9. Prescriptive Analytics
Having explored and evaluated the data available, Prescriptive analytics sits at the highest level of data analysis, as it focuses on recommending what should be done to optimize actions to achieve the desired outcomes. It evaluates multiple scenarios and constraints to identify the best decision (to promote optimization and efficiency).

10. Data Literacy.
The ability to read, understand, analyze, and communicate data in a meaningful manner.
e.g. A manager at Shoprite looks at a Sales dashboard.
- Sees a spike in sales for a product. → Asks: "Was there a promotion?"
- Checks the data and confirms it was a weekend promotion.

B. Data, Measurement & Reporting

11. Structured Data

This is data that is in a consistent format such as a table with rows and columns. Easy to store and use for analysis.

e.g. Sales records (transaction ID, Product, date, price); financial transactions.

12. Unstructured Data

This is data with no predefined format or structure. It ~~was~~ is more complex to store, process and analyze.

e.g. ~~Soci~~ Social media posts, emails, images, videos, etc..

13. Data Warehouse

A data warehouse is a large storage facility that stores multiple databases from various servers. It is a centralized storage system designed specifically for querying, reporting and analysis.

e.g. Data warehouse for retailer, Shoprite.

- Shoprite operates hundreds of stores * Point-of-sale systems (sales transactions) * Inventory systems * Customer loyalty programs.

All this data is stored in a data warehouse.

14. Data Mining

Using statistical methods, machine learning, and algorithms to ~~delve~~ dive deeper into the data within the large dataset to discover hidden patterns, relationships and useful insights.

e.g. Amazon uses data mining to understand and analyze customer purchase data to discover things such as; Customers who buy a phone often also buy a phone case and screen protector.

15. Data Visualisation

The process of representing data graphically using charts, graphs, maps and dashboards. This makes information easier to understand, interpret and communicate.

16. Dashboard

A visual display of key data, metrics, and performance indicators, organized in one place to provide a quick and clear overview of how a business or system is performing. It tracks KPI's and quickly identifies trends, issues or opportunities

17. Metric

A quantifiable measure used to track, assess and evaluate the performance of a process, activity or business objective.
eg. Revenue (total sales), Profit margin (%), conversion rate (%), etc..

18. Key Performance Indicator (KPI)

A measurable metric that is directly tied to a critical business objective and it is used to evaluate how effectively that goal is being achieved.
→ KPI is a key metric that matters most for success. All KPI's are metrics but not all metrics are KPI's.

19. Benchmark

Benchmarking is a strategic management tool used by organisations to compare their processes, products/services against industry standards or best practices.
→ Used as a comparative success measure.

20. Segmentation

Segmentation in Data is the process of using data analysis and algorithms to divide a dataset into distinct groups (segments) where the members within each group are similar to each other and different from those in other groups.

c. Customer & Sales KPI's

21. Conversion rate

The percentage of total visitors (potential client engaging with business/product) who purchase a product/service out of the total number of people who were exposed to or engaged with a process.

$$\text{Conversion rate} = \frac{\text{Number of Conversions}}{\text{Total visitors}} \times 100$$

22. Customer Acquisition Cost (CAC)

· The total average cost a business spends to acquire a new customer, including marketing, sales, and related expenses.

$$CAC = \frac{\text{Total Sales and Marketing Costs}}{\text{Number of New Customers Aquired}}$$

23. Customer Lifetime Value (CLV/LTV)

The total revenue a business expects to earn from a customer over the entire duration of their relationship with the company. The value of a customer to a business, over time.

24. Churn Rate

The percentage of customers (or users) who stop using a productor Service over a specific period of time. It measures customer loss and is a key indicator of how well a business retains its customers.

25. Customer Retention Rate

It measures how well a business is able to retain existing customers. It is the percentage of customers a business keeps over a specific period of time, excluding new customers gained during that period.

26. Net Promoter Score (NPS)
A metric used to measure customer loyalty and satisfaction by asking how likely customers are to recommend a company, product or service to others.

→ It is based on a simple survey question:
" ~~How~~ On a scale of 0 to 10, how likely are you to recommend us to a friend or colleague?".

27. Average Order Value (AOV)

$$AOV = \frac{\text{Total Revenue}}{\text{Number of Orders}}$$ → Measures the average amount spent per transaction.

(Torres. R, 2023).

28. Sales Funnel
A visual, step-by-step model that maps the journey a potential customer takes from ~~init~~ initial awareness to making a final purchase. Focuses on the conversion rates and volume of prospects at each stage of their journey.

D. Marketing & Campaign Analytics

29. Marketing Campaign
Marketing Campaigns are a series of strategic actions designed to Promote specific business products and goals executed within a set time frame.

30. Post-Campaign Analysis
The systematic process of evaluating the effectiveness, performance, and ROI of a marketing campaign after it has concluded. It involves gathering data from multiple channels to determine what worked, what didn't and ~~w~~ why, converting raw metrics into actionable insights.

31. Incremental revenue is the additional revenue your business generates directly because of the specific marketing campaign or promotion. It's the money generated (that otherwise wouldn't have) due to the campaign or promotion. (Caper, G, 2026).

32. Return on Investment (ROI)

$$ROI = \frac{\text{Net Profit}}{\text{Cost of Investment}} \times 100$$

This is a financial metric used to evaluate the efficiency and profitability of an investment. It acts as a KPI that measures how much income or value an investment generates relative to its expenses.

33. Return on Ad Spend (ROAS)

$$ROAS = \frac{\text{Revenue Generated from Advertising}}{\text{Cost of Advertising}}$$

- It is a fundamental KPI in digital marketing and business analytics that measures the gross revenue generated for every rand spent on advertising. "
- It determines financial efficiency of a specific ad campaign. (Google, 2026)

34. Click-Through Rate (CTR)

$$CTR = \frac{\text{Total Clicks}}{\text{Total Impressions (views)}} \times 100$$

A digital marketing metric measuring the percentage of people who click on a link (advertisement, email or search result) after seeing it.

35. Bounce Rate

The percentage of visitors who enter a website or app and leave after viewing only one page, without interacting with the site. It is a key indicator of user engagement and content relevance.

36. A/B Testing

Also known as split testing, is a randomized, controlled experiment used to compare two versions of a digital asset. A control A and Variant B to determine which performs better against a defined metric. i.e. conversion rate or user engagement.

37. Lead Generation

The strategic process of identifying, attracting, and nurturing potential customers (leads) who have shown interest in a product or service, ultimately converting them into paying customers.

38. Marketing Attribution.

- It is the practice of assigning credit for ~~conversation~~ conversions or revenue to marketing touchpoints in order to pinpoint the touchpoints and channels that are working best and allocate resources accordingly.

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